Cap Plasticity in LCM: Reference Formulation, Integration, and Verification

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Abstract

LCM implements a single cap-plasticity constitutive model, `CapModel` (material-database name `Cap`): the three-invariant isotropic/kinematic hardening cap model of the Sandia GeoModel lineage [3–5], in the form presented in Section 3 of Sun et al. [9]. This document states the formulation as implemented, including the conventions that the published papers leave ambiguous and that were settled against the original sources and the LAME GeoModel reference implementation; documents the adaptive substepped explicit integration; describes the logarithmic-strain / Kirchhoff-stress kinematics used for finite deformation; summarizes why no algorithmic tangent is ever hand-coded (Sacado automatic differentiation produces all of them); and describes the verification of the implementation against an independent reference implementation, which reproduces the kernel to machine precision over full load histories.

Keywords: Cap plasticity, three-invariant yield surface, kinematic hardening, pore collapse, adaptive substepping, logarithmic strain, Kirchhoff stress, automatic differentiation, verification, LCM.

1 Scope, Sources, and Conventions

`CapModel` lives in `src/LCM/models/CapModel.{hpp,cpp}` and `CapModel_Def.hpp` and is selected in the material database with `Model Name: Cap`. It supports small-strain and finite-deformation kinematics through the boolean material parameter `Finite Deformation` (Section 7); the constitutive integrator is identical in both cases.

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A fully implicit variant (`CapImplicit`, a 13-unknown local backward-Euler solve following Foster et al. 5) existed in LCM until 2026 and was removed: it was unmaintained, had no test coverage, and its principal classical advantage — the consistent algorithmic tangent — carries little weight in LCM, where automatic differentiation produces the exact tangent of whatever integrator actually runs (Section 8). The accuracy concern that remains for an explicit scheme, large strain increments, is addressed by adaptive substepping (Section 6). The model's own authors made the same choice, moving from the implicit formulation of Foster et al. [5] to the refined explicit scheme of Sun et al. [9] for production work.

Sources and their precedence. The implementation follows Sun et al. [9, Section 3]. Where that paper is ambiguous, in error, or silent, the conventions are settled by, in order: the original model papers [5, 6], and the LAME GeoModel reference implementation (`iso_geomodel_model.F`, the production Sandia code documented in Fossum and Brannon 3). Each such arbitration is called out where it appears below. Two errata in Sun et al. [9] as printed are relevant:

1. Algorithm 1, Step 2 prints the stress update as $\boldsymbol{\sigma}_{n+1} = \boldsymbol{\sigma}^{\mathrm{tr}} - \Delta\gamma\, \partial g/\partial\boldsymbol{\sigma}$, omitting the elastic tangent. The $\Delta\gamma$ derivation in the same paper, and Algorithm 2, both require $\boldsymbol{\sigma}_{n+1} = \boldsymbol{\sigma}^{\mathrm{tr}} - \Delta\gamma\, C^e\!:\!\partial g/\partial\boldsymbol{\sigma}$, which is what LCM implements.

2. Algorithm 2's normal-correction fallback mixes C^e into the stress update while omitting it from the denominator $\tilde{\chi} = \partial_\sigma f : \partial_\sigma f$, which is dimensionally inconsistent. LCM implements the consistent Sloan et al. [8] form: a pure geometric projection with no elastic tangent in either place (Section 6).

Sign convention. Tension positive, compression negative, throughout. Under confinement $I_1 = \operatorname{tr}\boldsymbol{\sigma} < 0$; the cap sits on the compressive (negative-I_1) side, and "the cap expands" means its hydrostat intercept moves toward $-\infty$.

2 Physical Overview

Geomaterials exhibit two inelastic mechanisms that the yield surface must represent simultaneously. At low to moderate pressure the dominant mechanism is frictional sliding and growth of microcracks: shear resistance grows with confinement, producing a pressure-dependent *shear failure envelope* that, unlike a Drucker–Prager cone, saturates once the cracks are closed by the confining pressure. At high pressure a porous material yields under hydrostatic load alone by irreversible *pore collapse*; representing this requires closing the yield surface on the compressive end with a *cap*, which moves outward as porosity is crushed away — the isotropic hardening of the model. Shear stress assists pore collapse, so the cap is an ellipse rather than a flat cutoff. Directionality of frictional sliding produces a strong Bauschinger effect, represented by a deviatoric back stress that translates the yield surface up to, but never beyond, the fixed failure envelope. Finally, geomaterials are weaker in triaxial extension than in triaxial compression, which brings the third stress invariant into the yield function. The classical two-surface cap models [e.g. 4] join envelope and cap at a corner; the present formulation is a single C^1 surface, which is the property that makes the explicit integration well behaved.

3 Kinematics in the Small-Strain Path

The small-strain path additively decomposes the strain rate [9, Eq. 47],

$$\dot{\boldsymbol{\varepsilon}} \;=\; \dot{\boldsymbol{\varepsilon}^e} + \dot{\boldsymbol{\varepsilon}^p}, \tag{1}$$

and uses isotropic linear elasticity [9, Eqs. 49–50],

$$\dot{\boldsymbol{\sigma}} \;=\; C^e : \dot{\boldsymbol{\varepsilon}^e} \;=\; C^e : (\dot{\boldsymbol{\varepsilon}} - \dot{\boldsymbol{\varepsilon}^p}), \qquad C^e \;=\; \lambda \boldsymbol{I} \otimes \boldsymbol{I} + 2\mu I, \tag{2}$$

with λ, μ the Lamé constants and I the symmetric fourth-order identity. Given state $\{\boldsymbol{\sigma}_n, \boldsymbol{\alpha}_n, \kappa_n\}$ and strain increment $\Delta\boldsymbol{\varepsilon}$ from the surrounding global Newton iteration, the trial elastic stress is

$$\boldsymbol{\sigma}^{\mathrm{tr}} \;=\; \boldsymbol{\sigma}_n + C^e : \Delta\boldsymbol{\varepsilon}. \tag{3}$$

4 Invariants, Lode Coefficient, and Yield Surface

Kinematic hardening is introduced through the deviatoric back stress $\boldsymbol{\alpha}$ via the relative stress

$$\boldsymbol{\xi} \;:=\; \boldsymbol{\sigma} - \boldsymbol{\alpha}, \tag{4}$$

with invariants

$$I_1 = \operatorname{tr}\boldsymbol{\xi}, \tag{5}$$

$$J_2^\xi = \tfrac{1}{2}\,\boldsymbol{s} : \boldsymbol{s}, \qquad \boldsymbol{s} \;=\; \boldsymbol{\xi} - \tfrac{1}{3} I_1 \boldsymbol{I}, \tag{6}$$

$$J_3^\xi = \det \boldsymbol{s}. \tag{7}$$

Because $\boldsymbol{\alpha}$ is deviatoric, $\operatorname{tr}\boldsymbol{\xi} = \operatorname{tr}\boldsymbol{\sigma}$, so the ξ superscript is used on J_2, J_3 only.

Lode coefficient. The difference in strength between triaxial compression and triaxial extension enters through [9, Eqs. 34–36]

$$\Gamma(\beta) \;=\; \tfrac{1}{2}\left[1 - \sin 3\beta + \frac{1}{\psi}\,(1 + \sin 3\beta)\right], \qquad \sin 3\beta \;=\; -\frac{3\sqrt{3}\,J_3^\xi}{2\,(J_2^\xi)^{3/2}}, \tag{8}$$

where $\psi \le 1$ is the ratio of triaxial-extension to triaxial-compression strength. In triaxial compression $\Gamma = 1$; in triaxial extension $\Gamma = 1/\psi$, and since Γ^2 multiplies J_2^ξ in the yield function, yield in extension occurs at ψ times the compressive strength. The code sets $\Gamma \equiv 1$ when $\psi = 0$ or $J_2^\xi = 0$ to avoid the singularity on the hydrostat. With $\psi = 1$ the Lode dependence is inactive and all J_3 terms drop out of the stress gradients — a fact relevant to the implementation history (Appendix A).

Shear failure surface, cap, and cap position. The pressure-dependent shear failure surface and the smooth elliptical hardening cap [9, Eqs. 32, 37–38] are

$$F_f(I_1) = A - C\exp(D\,I_1) - \theta\,I_1, \tag{9}$$

$$F_c(I_1,\kappa) = 1 - H(\kappa - I_1)\,\frac{(I_1-\kappa)^2}{[X(\kappa)-\kappa]^2}, \tag{10}$$

$$X(\kappa) = \kappa - R\,F_f(\kappa), \tag{11}$$

where H is the Heaviside function. The internal variable κ is the *branch point* where the cap begins, and $X(\kappa)$ is the *hydrostat intercept*: the current hydrostatic crush strength. On the shear branch $I_1 > \kappa$, $F_c \equiv 1$; on the compactive cap $X < I_1 \leq \kappa$, F_c falls quadratically from one to zero, closing the surface at $I_1 = X$. At $I_1 = \kappa$ both $F_c = 1$ and $\partial F_c/\partial I_1 = 0$, so the surface is C^1 across the branch point: there is no corner where cap meets envelope. The strength at zero pressure is $A - C$; as compression grows the exponential term vanishes and strength saturates toward $A - \theta I_1$. R is the cap aspect ratio (its extent in I_1 per unit of F_f), and N offsets the initial yield surface below the failure envelope, leaving room for kinematic hardening.

Yield function and plastic potential.

$$f(I_1, J_2^\xi, J_3^\xi, \boldsymbol{\alpha}, \kappa) = \Gamma^2 J_2^\xi - F_c(I_1,\kappa)\left(F_f(I_1) - N\right)^2, \tag{12}$$

$$g(I_1, J_2^\xi, J_3^\xi, \boldsymbol{\alpha}, \kappa) = \Gamma^2 J_2^\xi - F_c^g(I_1,\kappa)\left(F_f^g(I_1) - N\right)^2, \tag{13}$$

with the plastic-potential analogues

$$F_f^g(I_1) = A - C\exp(L\,I_1) - \phi\,I_1, \tag{14}$$

$$F_c^g(I_1,\kappa) = 1 - H(\kappa - I_1)\,\frac{(I_1-\kappa)^2}{[X^g(\kappa)-\kappa]^2}, \tag{15}$$

$$X^g(\kappa) = \kappa - Q\,F_f(\kappa). \tag{16}$$

Non-associativity ($L \neq D$, $\phi \neq \theta$, $Q \neq R$) exists because associated flow on a frictional envelope grossly over-predicts dilatancy; flattening the potential's pressure dependence ($\phi < \theta$, $L < D$) reduces the predicted plastic volume expansion to measured levels. Associative flow corresponds to $L = D$, $\phi = \theta$, $Q = R$.

Convention note. In Eq. (16) the potential cap position X^g is built from the *yield* failure function F_f (parameters D, θ), not from F_f^g: only the aspect ratio Q distinguishes the potential cap from the yield cap. This follows Regueiro and Foster [6, Eq. 14] and the LAME reference implementation. The distinction is invisible for associative parameters and matters otherwise; the pre-2026 LCM code had it wrong (Appendix A).

5 Flow Rule and Hardening Evolution

The plastic strain rate follows the gradient of the plastic potential [9, Eq. 48],

$$\dot{\boldsymbol{\varepsilon}}^p \;=\; \dot\gamma\,\frac{\partial g}{\partial \boldsymbol{\sigma}}, \tag{17}$$

where $\dot\gamma \geq 0$ is the consistency parameter.

Kinematic hardening. The deviatoric back stress evolves along the deviator of the flow direction [9, Eqs. 41–43],

$$\dot{\boldsymbol{\alpha}} \;=\; \dot\gamma\,\boldsymbol{h}^\alpha, \qquad \boldsymbol{h}^\alpha \;=\; c^\alpha\,G^\alpha(\boldsymbol{\alpha})\,\mathrm{dev}\!\left(\frac{\partial g}{\partial \boldsymbol{\sigma}}\right), \qquad G^\alpha \;=\; \max\!\left(0,\; 1-\frac{\sqrt{J_2^\alpha}}{N}\right), \tag{18}$$

with $J_2^\alpha = \frac{1}{2}\boldsymbol{\alpha} : \boldsymbol{\alpha}$. The limiter G^α drives the hardening rate to zero as the translating yield surface approaches the fixed failure envelope, which it reaches when $\sqrt{J_2^\alpha} = N$.

The clamp at zero in Eq. (18) is essential and absent from the published papers. The explicit update can overshoot $\sqrt{J_2^\alpha}$ slightly past N; without the clamp, $G^\alpha < 0$ *reverses* the direction of kinematic hardening. At a material point this is a small error, but at the structural level it destabilizes the homogeneous solution: in LCM's verification study a uniform-property confined-compression problem on a $2 \times 2 \times 2$ mesh demonstrably bifurcated to a non-homogeneous Newton solution at exactly the step where the back stress reached its limit surface. The LAME reference implementation has always floored its equivalent quantity (`GFUN`) at zero. See Appendix A.

Isotropic (cap) hardening. The cap position is driven by volumetric plastic strain through the experimentally measured crush curve [9, Eqs. 44–46],

$$\dot\kappa \;=\; \dot\gamma\,h^\kappa, \qquad h^\kappa \;=\; \frac{\mathrm{tr}(\partial g/\partial \boldsymbol{\sigma})}{\partial \varepsilon_v^p/\partial \kappa}, \qquad \varepsilon_v^p(\kappa) = W\left[e^{\,D_1\bar X - D_2 \bar X^2} - 1\right], \quad \bar X = X(\kappa) - X(\kappa_0), \tag{19}$$

where κ_0 is the initial cap branch point and W, D_1, D_2 are material parameters. W is the maximum compactive volumetric plastic strain, essentially the initial porosity: as $X \to -\infty$, $\varepsilon_v^p \to -W$ and all porosity has been crushed out. The denominator in h^κ is evaluated by the chain rule $(\partial \varepsilon_v^p/\partial X)(\mathrm{d}X/\mathrm{d}\kappa)$ in `compute_dedkappa`.

Convention note. The crush curve in Eq. (19) is parametrized by the *yield* cap position $X(\kappa)$ of Eq. (11) — parameters R, D, θ — because X is the measurable hydrostatic elastic limit against which the curve is calibrated. The LAME reference computes $\mathrm{d}X/\mathrm{d}\kappa = 1 - R\,\mathrm{d}F_f/\mathrm{d}\kappa$ accordingly. The pre-2026 LCM code used the potential-surface quantities (Q, L, ϕ) here, which is wrong for non-associative parameters (Appendix A).

Two safeguards from the LAME reference, absent from the papers, apply wherever κ is updated:

- *No cap contraction:* increments of κ are clamped to $\Delta\kappa \le 0$ (LAME: `HK = MIN(HK,0)`, comment "prevents cap contraction"). Dilation never re-opens porosity.

- *Cap lock at full compaction:* when $|\varepsilon_v^p(\kappa)| \ge W$ the crush curve is exhausted, its slope vanishes, and h^κ as written diverges. Following the reference, h^κ is then set to $-0.01\,K^2$ (K the bulk modulus), which drives the cap rapidly toward $-\infty$ so that pore-collapse plasticity shuts off rather than stalling against an immobile cap.

Strain softening by loss of cohesion. Frozen sand softens after its peak: in the Xu, Wu and Qi (2016) triaxial series at $-6\,°\mathrm{C}$ the deviatoric stress falls to 62, 70, 74 and 77 per cent of its peak by 26 per cent axial strain at $P_c = 3.0 \times 10^5$, 6.0×10^5, 8.0×10^5 and 1.0×10^6 Pa. Nothing above can produce that: the envelope F_f is fixed and kinematic hardening only moves the back stress inside it. The physical mechanism is ice-bond breakage under shear, and the data locate it precisely — the strength lost is 70 to 80 per cent of the *cohesive* share $A - C$ of the peak strength at every confining pressure, while the frictional share θI_1 is untouched.

The model follows the LAME Kayenta implementation [2], the production descendant of the GeoModel this cap model descends from, which carries a scalar *coherence* ("one minus damage") and blends an intact and a failed limit surface with it. Here the failed surface is the intact one with its cohesion reduced, so a single scalar $\omega \in [\omega_{\mathrm{res}}, 1]$ acts on the cohesive group alone:

$$A - C \;\to\; \omega\,(A - C), \qquad N \;\to\; \omega\,N, \tag{20}$$

implemented as $C \to A - \omega(A - C)$ so that A, θ, the envelope curvature Ce^{DI_1} and every cap parameter are left as they are. This is the same algebra the Permafrost model applies for ocean-exposure weakening. When N shrinks, a back stress that had reached the old N is scaled back onto the new bounding surface $\sqrt{J_2^\alpha} \le N$, so that the reach of the translated yield surface, $(F_f - N) + \sqrt{J_2^\alpha}$, is the softened envelope and not more; the clamp in Eq. (18) only stops growth.

Damage begins when the kinematic hardening is exhausted, $\sqrt{J_2^\alpha} \ge (1 - 10^{-6})\,N$ (Kayenta: `GFUN` $\le 10^{-6}$), and accumulates only on the shear branch $I_1 \ge \kappa$ (Kayenta: `.NOT.CONFINED`); nothing softens while the material is still hardening or while it is on the compaction cap, so the design rule of Section 12 that environmental softening never flows through κ holds here too. The driver is the equivalent plastic strain accumulated after onset, s, and the law is Kayenta's logistic in that strain,

$$\omega(s) \;=\; \omega_{\mathrm{res}} + \Big(1 - \omega_{\mathrm{res}}\Big)\,\frac{1 + e^{-2k}}{1 + e^{-2k\,(1 - s/\varepsilon_f)}}, \tag{21}$$

which is 1 at $s = 0$, halfway to the residual at $s = \varepsilon_f$ (`Failure Strain`), and tends to ω_{res} (`Coherence Residual`) with a sharpness set by k (`Failure Speed`: Kayenta's own guidance is that ~ 0.1 is gradual and ~ 30 abrupt). Coherence never heals. The coherence converged at the previous step is applied to the whole of the current step, the same one-step lag the Permafrost map uses for ε_v^p; with several hundred steps per test the lag is invisible. Two states carry it, `Coherence` and `Damage_Strain`, and both kernels share the code in `CapSoftening.hpp`. The reference implementation mirrors it, and `CapModel_Verify_softening` and `Permafrost_Verify_softening` compare the full coherence history on a deviatoric path (axial compression with lateral extension, since confined compression of Salem limestone never leaves the elastic range on the shear branch) to 2×10^{-15}.

Two things Kayenta does that this deliberately does not: reduce the elastic moduli by the coherence (nothing in a monotonic triaxial test constrains it), and drive damage by time rather than strain (rate dependence through a crack growth time, the natural hook when creep comes up). And one thing neither does: a softening material point is well posed, a softening continuum is not — a boundary-value problem with this law localizes and depends on its mesh until a length scale is introduced. That regularization is a separate decision.

6 Integration: Substepped Explicit Scheme

The integrator is the refined explicit scheme of Sun et al. [9, Algorithms 1–2], upgraded with the adaptive substepping of Sloan et al. [8]: within the plastic part of a strain increment, modified-Euler (RK1/RK2) pairs advance the state, the difference between the two estimates supplies a local error measure that controls substep size, and every accepted substep is followed by a drift correction back to the yield surface.

Rate evaluation. At state $\{\boldsymbol{\sigma}, \boldsymbol{\alpha}, \kappa\}$, for a substrain $\Delta\boldsymbol{\varepsilon}_{\text{sub}}$, the consistency condition $\dot{f} = 0$ gives

$$\Delta\gamma \;=\; \frac{(\partial f/\partial\boldsymbol{\sigma}) : C^e : \Delta\boldsymbol{\varepsilon}_{\text{sub}}}{\chi}, \qquad \chi \;=\; \tfrac{\partial f}{\partial \boldsymbol{\sigma}} : C^e : \tfrac{\partial g}{\partial \boldsymbol{\sigma}} - \tfrac{\partial f}{\partial \boldsymbol{\alpha}} : \boldsymbol{h}^{\alpha} - \tfrac{\partial f}{\partial \kappa}\, h^{\kappa}, \tag{22}$$

with $\partial f/\partial\boldsymbol{\alpha} = -\partial f/\partial\boldsymbol{\sigma}$. $\Delta\gamma$ is clamped at zero (elastic unloading within a substep), and the increments are

$$\Delta\boldsymbol{\sigma} = C^e : \Delta\boldsymbol{\varepsilon}_{\text{sub}} - \Delta\gamma\, C^e : \tfrac{\partial g}{\partial\boldsymbol{\sigma}}, \qquad \Delta\boldsymbol{\alpha} = \Delta\gamma\, \boldsymbol{h}^{\alpha}, \qquad \Delta\kappa = \min(0,\, \Delta\gamma\, h^{\kappa}). \tag{23}$$

Algorithm. With input state $\{\boldsymbol{\sigma}_n, \boldsymbol{\alpha}_n, \kappa_n\}$ and strain increment $\Delta\boldsymbol{\varepsilon}$:

1. *Elastic trial* via (3). If $f(\boldsymbol{\sigma}^{\text{tr}}, \boldsymbol{\alpha}_n, \kappa_n) \le 0$, accept the trial and stop.
2. *Substepping*, starting from the converged state (not the trial), with pseudo-time $T \in [0, 1]$, $\Delta T = 1$ initially, and $\Delta T_{\min} = 1/$`Maximum Substeps`. While $T < 1$:
   (a) Evaluate the increments (23) at the current state for $\Delta\boldsymbol{\varepsilon}_{\text{sub}} = \Delta T\, \Delta\boldsymbol{\varepsilon}$ (RK1), and again at the RK1-advanced state (RK2).
   (b) Form the modified-Euler state from the average of the two increment sets, and the relative error estimate

$$\eta \;=\; \frac{\|\Delta\boldsymbol{\sigma}_2 - \Delta\boldsymbol{\sigma}_1\|}{2\, \max(\|\boldsymbol{\sigma}_{\text{new}}\|,\, \epsilon)}. \tag{24}$$

(c) If $\eta >$ `Substep Tolerance` and $\Delta T > \Delta T_{\min}$: reject, set $\Delta T \leftarrow \max\Big(0.9\sqrt{\mathrm{STOL}/\eta},\, 0.1\Big)\,\Delta T$ (floored at $\Delta T_{\min}$), retry.

(d) Otherwise accept: apply the drift correction below, advance $T \leftarrow T + \Delta T$, and grow $\Delta T \leftarrow \min\Big(0.9\sqrt{\mathrm{STOL}/\eta},\, 1.1\Big)\,\Delta T$, capped at $1 - T$.

3. *Plastic strain increment* recovered through elastic compliance,

$$\Delta\boldsymbol{\varepsilon}^p \;=\; (C^e)^{-1} : (\boldsymbol{\sigma}^{\mathrm{tr}} - \boldsymbol{\sigma}_{n+1}), \tag{25}$$

from which the outputs ε_v^p += $\operatorname{tr}\Delta\boldsymbol{\varepsilon}^p$ and $\varepsilon_{\mathrm{eqps}}$ += $\sqrt{\frac{2}{3}}\,\|\operatorname{dev}\Delta\boldsymbol{\varepsilon}^p\|$ are accumulated.

Drift correction. Explicit updates do not satisfy $f = 0$ exactly. After each accepted substep, while $|f| > f_{\mathrm{tol}}$ and fewer than 20 iterations:

1. *Consistent correction:* $\delta\gamma = f/\chi$ with all quantities at the current state; update $\boldsymbol{\sigma} \leftarrow \boldsymbol{\sigma} - \delta\gamma\, C^e : \partial g/\partial\boldsymbol{\sigma}$, $\boldsymbol{\alpha} \leftarrow \boldsymbol{\alpha} + \delta\gamma\,\boldsymbol{h}^{\alpha}$, $\kappa \leftarrow \kappa + \min(0, \delta\gamma\, h^{\kappa})$.

2. *Normal-correction fallback:* if the consistent correction increases $|f|$, discard it and instead project geometrically,

$$\delta\gamma \;=\; \frac{f}{\partial_\sigma f : \partial_\sigma f}, \qquad \boldsymbol{\sigma} \leftarrow \boldsymbol{\sigma} - \delta\gamma\,\frac{\partial f}{\partial\boldsymbol{\sigma}}, \tag{26}$$

holding $\boldsymbol{\alpha}, \kappa$ fixed. This is the Sloan et al. [8] form; the version printed in Sun et al. [9] is dimensionally inconsistent (Section 1).

The yield function has units of stress squared, so the drift tolerance is scaled to make the convergence test independent of the unit system:

$$f_{\mathrm{tol}} \;=\; 10^{-12}\,E^2. \tag{27}$$

(The historical absolute tolerance of 10^{-10} was unreachable in Pa-based unit systems, so the corrector always ran to its iteration cap.)

Branch consistency under AD. The substep control flow (acceptance, rejection, step growth) branches on Sacado *values*, so the Residual and Jacobian evaluations of the same state take identical branch sequences and the AD tangent differentiates exactly the code path that produced the stress.

Evolving material parameters. The integrator accepts *two* parameter sets, for the beginning and end of the strain increment, and ramps linearly between them across the substep pseudo-time: RK1 rates use the substep-start parameters, RK2 rates and the drift correction the substep-end parameters. With constant parameters (the `Cap` model) the two sets are equal and the scheme reduces to the fixed-parameter integrator. This mechanism exists for models whose parameters follow an external field — the Permafrost model's ice saturation — and it is not merely an accuracy refinement. If instead the parameters jump once per step, a softening step leaves the stored stress far outside the shrunken yield surface, and the entire correction is delivered by the drift projection, whose consistent/normal fallback decision then becomes critically sensitive: in a thaw-under-load verification study the resulting trajectories were hypersensitive to roundoff (relative perturbations of 10^{-16} in the moduli grew to order-one trajectory differences, with two distinct attractors). Ramping the parameters keeps each substep's surface motion small, the drift corrections small, and the trajectory robust — the same study then matched the reference implementation to $\sim 10^{-15}$.

Two consistency requirements accompany evolving parameters. First, the elastic strain, not the stress, is the state: when the moduli change between steps, the stored stress must be re-expressed through the current stiffness, $\boldsymbol{\sigma}_n \leftarrow C^e : C^{e\,-1}_{\text{old}} : \boldsymbol{\sigma}_n$, before integrating (the finite-deformation path does this by construction, recovering the elastic logarithmic strain from the stored plastic geometry). Second, the previous step's parameter state must be available; the Permafrost model stores its ice saturation as a state variable for this purpose.

Material parameters of the integrator. `Substep Tolerance` (STOL, default 10^{-4}) and `Maximum Substeps` (default 200). A rejected attempt counts toward an overall attempt cap of 4×`Maximum Substeps`.

Hand-coded derivatives. The closed-form constitutive derivatives $\partial f/\partial\boldsymbol{\sigma}$, $\partial g/\partial\boldsymbol{\sigma}$, $\partial f/\partial\kappa$, $\boldsymbol{h}^\alpha$, and $\partial\varepsilon_v^p/\partial\kappa$ are provided in `CapModel_Def.hpp` (`compute_dfdsigma`, `compute_dgdsigma`, `compute_dfdkappa`, `compute_halpha`, `compute_dedkappa`). The only nontrivial tensor derivative among them is

$$\frac{\partial J_3}{\partial\boldsymbol{\sigma}} \;=\; \boldsymbol{s}\cdot\boldsymbol{s} - \tfrac{2}{3}J_2\,\boldsymbol{I}, \tag{28}$$

a matrix product, *not* the elementwise square of $\boldsymbol{s}$; the two coincide only in principal axes (Appendix A). Every one of these closed forms is checked against complex-step differentiation by the verification suite (Section 9).

7 Finite-Deformation Kinematics

With `Finite Deformation: true` the model uses the multiplicative split and the isotropic logarithmic-strain / Kirchhoff-stress framework [7]: because every constitutive function in Sections 4–5 is an isotropic function of its tensor arguments, the small-strain integrator of Section 6 runs *unchanged* with the Kirchhoff stress $\boldsymbol{\tau}$ and the logarithmic elastic strain in the slots that otherwise hold $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$. Only the kinematic pre- and post-processing differ. At each quadrature point, with incoming $\boldsymbol{F}$, stored $\boldsymbol{F^p}_n$, and the deformation gradient $\boldsymbol{F}_n$ of the previous converged step:

1. *Trial elastic left Cauchy–Green tensor* from the stored plastic configuration ($\boldsymbol{C}_p^{-1} =$

$(\boldsymbol{F}^p{}_n)^{-1}(\boldsymbol{F}^p{}_n)^{-\mathrm{T}})$,

$$\boldsymbol{b}^{\mathrm{etr}} \;=\; \boldsymbol{F}\,\boldsymbol{C}_p^{-1}\,\boldsymbol{F}^{\mathrm{T}}, \qquad \boldsymbol{\varepsilon}^{\mathrm{etr}} \;=\; \tfrac{1}{2}\,\log_{\mathrm{sym}}\boldsymbol{b}^{\mathrm{etr}}, \tag{29}$$

with $\log_{\mathrm{sym}}$ the principal (spectral) logarithm.

2. *Starting stress and strain increment.* The elastic logarithmic strain at t_n is recovered from the same plastic configuration and the old deformation gradient,

$$\boldsymbol{\varepsilon}^e{}_n \;=\; \tfrac{1}{2}\,\log_{\mathrm{sym}}\!\Big(\boldsymbol{F}_n\,\boldsymbol{C}_p^{-1}\,(\boldsymbol{F}_n)^{\mathrm{T}}\Big), \qquad \boldsymbol{\tau}_n = C^e : \boldsymbol{\varepsilon}^e{}_n, \qquad \Delta\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^{\mathrm{etr}} - \boldsymbol{\varepsilon}^e{}_n. \tag{30}$$

The integrator then starts from $\boldsymbol{\tau}_n$ with increment $\Delta\boldsymbol{\varepsilon}$, so its trial stress $\boldsymbol{\tau}_n + C^e : \Delta\boldsymbol{\varepsilon} = C^e : \boldsymbol{\varepsilon}^{\mathrm{etr}}$ is *exact* for the hyperelastic logarithmic model: the incremental form introduces no approximation in the elastic relation.

3. *Return mapping* of Section 6 on $\boldsymbol{\tau}$, yielding $\boldsymbol{\tau}_{n+1}, \boldsymbol{\alpha}_{n+1}, \kappa_{n+1}$.

4. *Plastic configuration update.* The new elastic logarithmic strain follows from the returned stress, $\boldsymbol{\varepsilon}^e{}_{n+1} = (C^e)^{-1} : \boldsymbol{\tau}_{n+1}$, whence

$$\boldsymbol{b}^e{}_{n+1} = \exp_{\mathrm{sym}}(2\,\boldsymbol{\varepsilon}^e{}_{n+1}), \qquad \boldsymbol{C}_{p,n+1}^{-1} = \boldsymbol{F}^{-1}\,\boldsymbol{b}^e{}_{n+1}\,\boldsymbol{F}^{-\mathrm{T}}, \qquad \boldsymbol{F}^p{}_{n+1} = \Big(\boldsymbol{C}_{p,n+1}\Big)^{1/2}, \tag{31}$$

where the last expression is the symmetric positive-definite square root. The rotational part of $\boldsymbol{F}^p$ is unobservable in an isotropic model, so the symmetric root is a legitimate representative; no exponential map of the flow direction is formed explicitly — it is absorbed in the spectral log/exp of $\boldsymbol{b}^e$.

5. *Cauchy stress* for assembly, $\boldsymbol{\sigma}_{n+1} = \boldsymbol{\tau}_{n+1}/J$ with $J = \det \boldsymbol{F}$.

The stored state in this mode is $\boldsymbol{F}^p$ (identity-initialized) plus the deformation gradient history that `MechanicsProblem` already maintains; the small-strain mode stores the strain tensor instead. All other state (stress, back stress, κ, $\varepsilon_{\mathrm{eqps}}$, ε_v^p) is common.

This wrapper is independent of how the inner return mapping is integrated, which is why the desire for finite-deformation support had no bearing on the decision to remove the implicit integrator.

8 Tangents Are Not Hand-Coded: Automatic Differentiation

Sun et al. [9] do not document a consistent algorithmic tangent $\partial\boldsymbol{\sigma}_{n+1}/\partial\boldsymbol{\varepsilon}_{n+1}$, nor a finite-deformation counterpart. LCM does not need either. LCM's evaluators are templated on an evaluation trait; for `PHAL::AlbanyTraits::Jacobian` every field entering the kernel — strain or deformation gradient, stored state, every intermediate of the return mapping — carries a Sacado forward-mode AD type whose active directions are the element-level global degrees of freedom. The entire integrator (trial, substepping, drift correction, finite-deformation pre/post-processing) executes in AD arithmetic, and the derivative of the output stress with respect to every element DOF emerges as a by-product. Because the substep control flow branches on Sacado values (Section 6), the differentiated path is exactly the primal path. Sensitivities with respect to material parameters (for calibration and inverse problems) are generated by the same mechanism. The only hand-coded derivatives in the model are the *constitutive* ones listed in Section 6 — part of the model's definition, not of its integrator.

9 Verification

The implementation is verified, not merely regression-tested, by the infrastructure in `tests/LCM/CapModelPlasticity3D`:

Independent reference implementation. `cap_reference.py` is a deliberately separate transcription of the model from the publications — not a port of the C++. It evaluates all stress gradients by complex-step differentiation (machine-exact for the analytic branches), so the C++ closed-form derivatives and the reference share no code and no derivation. Its self-check verifies, among other identities, the J_3 derivative (28) against complex step at a generic (non-principal-axes) state. Both errors are measured relative to the closed form's own scale, so the check means the same thing in any unit system: the matrix-product form agrees to $O(10^{-15})$, while the elementwise-square form is wrong by $O(1)$.

Material-point exactness of the test problems. Each verification problem is a single hexahedral element whose 24 degrees of freedom are all prescribed by face boundary conditions. With no free degrees of freedom, the finite-element problem *is* the material point: no equilibrium iteration intervenes between the prescribed strain history and the constitutive response, and the comparison with the reference is exact rather than approximate. (The importance of this is not academic: on a $2 \times 2 \times 2$ mesh the same problem bifurcated away from the homogeneous solution when the unclamped G^α defect of Appendix A was present.)

Load paths and results. Five paths are compared state-by-state (stress components, κ, ε_v^p) over their full histories against the reference, which runs the identical substepping algorithm:

Test	Path	Exercises	Max. rel. diff.
`hydrostatic`	$\boldsymbol{\varepsilon} = -0.02\,t\,\boldsymbol{I}$	cap only; analytic identities	$\sim 10^{-14}$
`confined`	$\varepsilon_{xx} = -0.04\,t$, 200 steps	shear–cap interaction, $\boldsymbol{\alpha}$ to limit	$\sim 10^{-15}$
`triaxial`	unequal normal strains	$\psi = 0.8$ (Lode), non-associative	$\sim 10^{-15}$
`confined_coarse`	confined, 20 steps	substep rejection/regrowth	$\sim 10^{-15}$
`confined_fd`, `triaxial_fd`	as above, finite deformation	log/exp kinematics	$\sim 10^{-12}$

The hydrostatic path additionally checks two analytic identities: on the hydrostat the converged stress must satisfy $I_1 = X(\kappa)$ (observed to $\sim 10^{-13}$), and the accumulated ε_v^p must follow the crush curve of Eq. (19) (observed to the per-step consistency error of the discretization). The finite-deformation comparisons carry the slightly larger $\sim 10^{-12}$ floor of the eigendecomposition round-trips. The comparisons require `python3` with `numpy` and `netCDF4` and are skipped with a status message when these are unavailable.

10 Material Parameters

All parameters are read from the material database with the names below. The model is unit-agnostic: parameters must be supplied in one consistent unit system, with the stress unit denoted σ in the table. The convention throughout LCM is base SI, with magnitudes written in scientific notation rather than as prefixed units: stresses in Pa, so 2.2547×10^{10} Pa and not 22547 MPa. The ACE permafrost decks, the verification decks of Section 9 and the MatCal/Dakota calibration harness in `tools/calibration` all follow it, so a parameter set transfers between them without rescaling. The Salem limestone values below are the associative set of Sun et al. [9, Table 1], converted from the MPa of that table (stress-like values $\times 10^6$, $1/\sigma$ values $\times 10^{-6}$, $1/\sigma^2$ values $\times 10^{-12}$).

Input name	Symbol	Salem	Units	Meaning
`A`	A	6.892e8	σ	saturated shear strength
`C`	C	6.752e8	σ	strength deficit at zero pressure ($A - C$ = cohesion-like intercept)
`D`	D	3.94e-10	σ^{-1}	pressure scale of the exponential, yield surface
`theta`	θ	0.0	—	linear pressure slope, yield surface
`N`	N	6.0e6	σ	offset of initial yield below failure envelope
`R`	R	28.0	—	cap aspect ratio, yield surface
`kappa0`	κ_0	-8.05e6	σ	initial cap branch point
`W`	W	0.08	—	maximum compactive volumetric plastic strain ($\approx$ porosity)
`D1`	D_1	1.47e-9	σ^{-1}	crush-curve shape (linear)
`D2`	D_2	0.0	σ^{-2}	crush-curve shape (quadratic)
`calpha`	c^α	1.0e11	σ	kinematic hardening rate
`psi`	ψ	1.0	—	triaxial extension/compression strength ratio
`L`	L	3.94e-10	σ^{-1}	pressure scale, plastic potential
`phi`	ϕ	0.0	—	linear pressure slope, plastic potential
`Q`	Q	28.0	—	cap aspect ratio, plastic potential
`Softening`	—	false	—	cohesion softening on/off, Eq. (21)
`Coherence Residual`	ω_{res}	*required*	—	coherence at full damage, $(0, 1]$
`Failure Strain`	ε_f	*required*	—	damage strain at half loss
`Failure Speed`	k	*required*	—	sharpness of the fall
`Substep Tolerance`	STOL	10^{-4}	—	relative stress-error target per substep
`Maximum Substeps`	—	200	—	smallest substep is 1/this fraction of the increment
`Finite Deformation`	—	false	—	log/Kirchhoff kinematics of Section 7

For orientation with the Salem values: the initial hydrostatic crush strength is $X_0 = \kappa_0 - R\,F_f(\kappa_0) \approx -4.6 \times 10^8$ Pa, i.e. pore collapse under pure pressure begins near $p \approx 1.53 \times 10^8$ Pa, while the cap branch point sits at only 8×10^6 Pa — a long, flat cap, typical for porous rock.

11 Calibration

The parameters of Section 10 are obtained from a small suite of standard rock-mechanics experiments. The groups below are listed in the order in which they should be calibrated, because later groups depend on earlier ones; the definitive fitting procedures are documented in Fossum and Brannon [3, Section 4]. All stress-like parameters must be supplied in one consistent unit system (D, L, D_1 carry units of 1/stress, D_2 of 1/stress^2, c^α of stress; W, ψ, R, Q are dimensionless), and by the convention of Section 10 that system is base SI: fit the data in Pa rather than converting the fitted parameters afterwards, since a curve in MPa is indistinguishable from one in Pa to any fitting procedure and simply yields stress-like parameters 10^6 too small. The automated route through MatCal and Dakota (`tools/calibration`, see its `README.md`) applies the same groups and the same ordering.

1. Elasticity (E, ν). From *unload–reload loops*, not initial loading slopes: first loading of a geomaterial includes crack closure and seating, which read as artificially soft moduli. The bulk modulus comes from unloading slopes of the hydrostatic test (pressure vs. volumetric strain), the shear modulus from unloading slopes of the triaxial shear leg. Note that the original GeoModel uses pressure-dependent elastic moduli while this implementation is linear elastic: calibrate (E, ν) at the pressure range of the intended application, and do not expect a single pair to span both the low- and high-confinement regimes of the same rock.

2. Shear failure envelope (A, C, D, θ). From *triaxial compression (TXC) tests at several confining pressures*: each test contributes one peak stress point $(I_1, \sqrt{J_2})$, and the collection across pressures traces the envelope (9) for a least-squares fit. The parameters have distinct anchors, which is what makes the fit well posed: $A - C$ is the intercept at $I_1 = 0$, anchored by the unconfined compression strength ($\sqrt{J_2} = \sigma_c/\sqrt{3}$ at $I_1 = -\sigma_c$); A is the high-pressure asymptote, anchored by the highest-confinement tests; $1/D$ is the I_1 scale of the transition between the two, requiring tests that span the curved part of the envelope; and θ is a residual linear slope at high pressure that should be held at zero unless the high-pressure data clearly keep climbing — it is strongly correlated with A and D when the tested pressure range is narrow.

3. Lode strength ratio (ψ). The ratio of triaxial-extension to triaxial-compression strength at the same mean stress, requiring *triaxial extension* tests (axial stress reduced below the confinement). These are harder to run, so ψ is often the first parameter omitted; $\psi = 1$ switches the Lode dependence off entirely. Convexity of the Gudehus-type $\Gamma(\beta)$ in Eq. (8) requires $\psi \geq 7/9 \approx 0.778$; do not fit below that bound.

4. Cap and crush curve (κ_0, W, D_1, D_2, R). One *hydrostatic crush test*, pressurized well past the elastic limit with periodic unload–reload loops to separate elastic from plastic volumetric strain, supplies most of this group. The elastic-limit point gives X_0 ($I_1 = -3p$ at the first departure from elastic response: the onset of pore collapse). Beyond it, the stress state sits at $I_1 = X(\kappa)$ whenever it yields, so the plot of plastic volumetric strain against I_1 *is* the crush curve of Eq. (19): W is its plateau — the total crushable volume fraction, which should agree with independently measured porosity (for Salem limestone, $W = 0.08$

matches 8 % porosity); D_1 sets the initial compaction rate; and D_2 is unidentifiable unless the measured curve is strongly non-exponential — keep it at zero otherwise.

The cap aspect ratio R requires yield points on the cap *away* from the hydrostat: TXC tests at high confinement in which the specimen compacts during shear (shear-enhanced compaction). The locus of compaction-onset points in the $(I_1, \sqrt{J_2})$ plane traces the cap ellipse, and R is its width-to-height ratio. Without such data, R is commonly carried over from similar rock (it varies less than the other parameters; roughly 15–30 for porous rocks).

κ_0 is then *derived, not measured*: invert $X(\kappa_0) = \kappa_0 - R\,F_f(\kappa_0)$ for κ_0 given the measured X_0 and the fitted R and F_f. A common input error is to place the measured crush pressure directly into `kappa0`; for Salem limestone the two differ by a factor of about 57 ($\kappa_0 = -8.05 \times 10^6$ vs. $X_0 \approx -4.6 \times 10^8$ Pa).

5. Hardening (N, c^α). Both come from the *pre-peak portion* of the TXC stress–strain curves — the same tests as group 2, using the curve shape rather than the peak. N is the offset in $\sqrt{J_2}$ between initial yield (departure from linearity, onset of acoustic emission, or first unload–reload hysteresis) and peak; the model assumes this offset is pressure independent, which should be checked across the test suite. c^α is fit to the curvature of the hardening branch between initial yield and peak: large values (Salem's 10^{11} Pa) drive the yield surface to the failure envelope almost immediately, so the response is nearly elastic–perfectly-plastic; small values give gradual hardening. From monotonic data alone N and c^α are partially degenerate (a larger offset traversed faster resembles a smaller offset traversed slower); they separate cleanly only with load-reversal (cyclic) data, where the back stress reveals itself through the reduced reverse-yield stress. If the application never reverses the loading, the degeneracy is harmless.

6. Non-associative flow (L, ϕ, Q). These parameters do not affect where yield occurs, only the direction of plastic flow, so they are calibrated from *plastic strain ratios* rather than stress states. L and ϕ control the volumetric component of flow on the shear branch — dilatancy: measure the dilation rate $\mathrm{d}\varepsilon_v^p/\mathrm{d}\varepsilon_q^p$ during post-yield shear at several pressures and fit $L < D$, $\phi < \theta$ so that $\mathrm{tr}(\partial g/\partial \boldsymbol{\sigma})$ matches. Q does the same on the cap, from the ratio of compactive volumetric to deviatoric plastic strain during shear-enhanced compaction. If volumetric plastic-strain data are unavailable, set the flow associative ($L = D$, $\phi = \theta$, $Q = R$) with the understanding that dilatancy will be overpredicted. Recall from Eq. (16) that $X^g = \kappa - Q\,F_f(\kappa)$ uses the *yield* F_f, so Q is a pure aspect-ratio parameter — it does not pull L or ϕ into the cap position.

Workflow and checks. The dependency order is: elasticity $\to$ failure envelope and $\psi \to$ crush curve $\to R \to \kappa_0$ (derived) $\to N, c^\alpha \to L, \phi, Q$. Constraints worth verifying on any fitted set: $A - C > N > 0$ (positive initial strength at zero pressure); $D, D_1, W, R, Q > 0$; $\kappa_0 < 0$; and $7/9 \le \psi \le 1$.

Two practical points for carrying this out in LCM. First, the verification decks of Section 9 are single-element, fully strain-controlled drivers: `input_hydrostatic.yaml` *is* a hydrostatic crush simulation and `input_confined.yaml` *is* an oedometer test, so copying one and editing the boundary-condition ramp yields a calibration driver that can be compared against each laboratory curve directly, with no structural effects in the way. Second, `Substep Tolerance` and `Maximum Substeps` are numerical parameters that require no calibration and should be left at their defaults; if results change when the tolerance is tightened, the global time step was too coarse — the remedy is not in the material parameters.

12 The Permafrost Model

`Permafrost` (material-database name `Permafrost`) is the cap-plasticity model specialized for frozen and thawing sediment in the ACE Arctic coastal-erosion application [1], and replaces the `J2Erosion` model there. `J2Erosion` extends J_2 plasticity with an ice-saturation–dependent yield stress ($\sigma_0 = \sigma^{\text{sed}} + S\,\sigma^{\text{ice}-\text{sed}}$, with $S \in [0, 1]$ the ice saturation from the sediment freezing curve) and, over time, accumulated five failure criteria and several empirical adjustments (polynomial fits of stiffness and strength in ice saturation and porosity, ocean-exposure weakening factors, a peat-dependent strain limit). `Permafrost` replaces those mechanisms with cap-model quantities. The integration and constitutive functions are the shared, verified integrator of Section 6; the kernel supplies parameters that vary per integration point with the ice saturation.

Why the cap model fits permafrost. The eroding material spans a continuum from ice-bonded (frozen) to thawed sediment, and the cap model contains both end members as parameter limits. With $D \to 0$ the failure envelope flattens to the pressure-insensitive strength $A - C$, and with $\psi = 1$ and the cap pushed far out the model degenerates to von Mises — the `J2Erosion` ice model. At the other extreme, thawed sediment is the material the cap model was designed for: frictional shear strength plus compaction of a porous skeleton, neither of which J_2 represents (in `J2Erosion` the thawed state is a constant residual yield stress). Two further physical arguments: frozen soils exhibit a closed strength envelope (strength rises with confinement, peaks, and falls at high pressure through pressure melting of pore ice), which the cap closure can represent phenomenologically and a J_2 cylinder cannot; and thaw settlement is compaction — crush-curve physics, with W supplied by the effective-porosity profiles the ACE model already carries.

The parameter map. Ice saturation enters the cap model in exactly one place: as a per–integration–point modulation of the *material parameters*. The kinematics (Section 3), the invariants and yield surface (Section 4), and the flow and hardening laws (Section 5) are used verbatim; no governing equation acquires an explicit saturation term. This is the same device as temperature–dependent yield in metal plasticity — the constitutive structure is fixed and the environment acts only through the coefficients. Let $S \in [0, 1]$ be the ice saturation, with $S = 1$ fully frozen and $S = 0$ fully thawed. At each integration point the kernel evaluates every cap parameter as a function of S and hands the resulting parameter set to the shared integrator of Section 6.

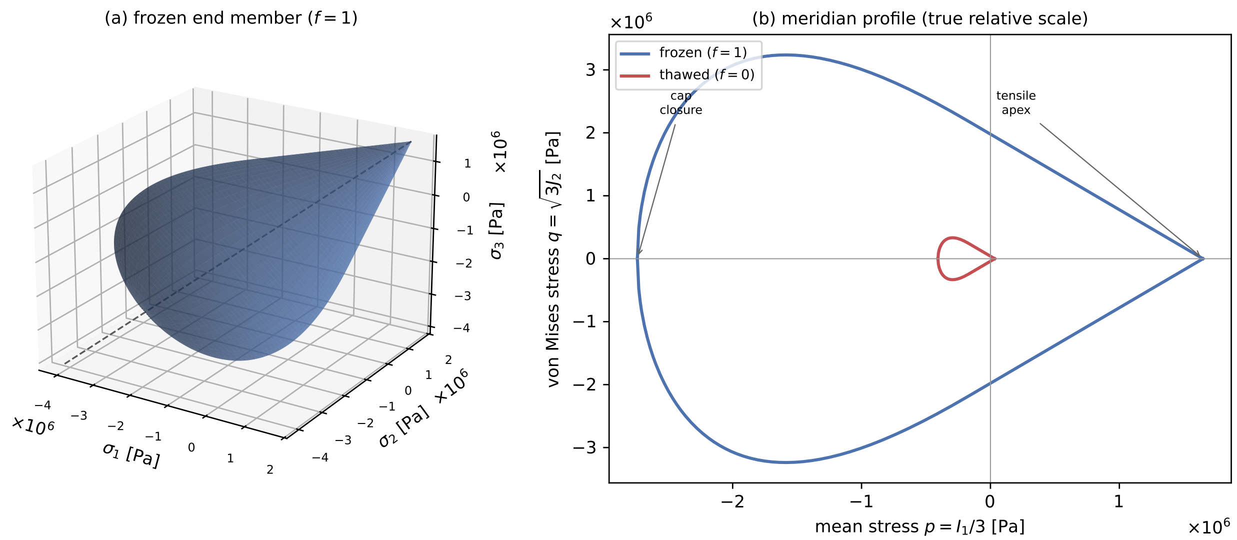

Figure 1: The `Permafrost` yield surface $f = \Gamma^2 J_2^{\xi} - F_c(I_1, \kappa)\,(F_f(I_1) - N)^2 = 0$ for the calibrated frozen and thawed end members (Eqs. (9)–(12), parameters in the text). Here $\psi = 1$ and $D \approx 0$, so the surface is a near-linear Drucker–Prager cone of circular cross-section closed by an elliptical cap — a body of revolution about the hydrostatic axis (dashed). **(a)** The frozen surface ($S = 1$) in principal-stress space. **(b)** Meridian profiles $q = \sqrt{3J_2}$ against $p = I_1/3$ for both end members on common axes: the cone opens from the tensile apex ($F_f = N$, at $p \approx 1.65 \times 10^6$ and 3.3×10^4 Pa) toward compression and the cap closes at the hydrostat intercept $X(\kappa)$ ($p \approx -2.74 \times 10^6$ and -4.1×10^5 Pa). Thaw shrinks the surface by about the cohesion ratio ($\approx 50\times$ here) while preserving its shape — the parameter map below, evaluated at its two ends. Generated by `figures/gen-yield-surface.py`.

The material input carries two calibrated end-member parameter sets, the `Frozen Parameters` and `Thawed Parameters` sublists. Two variables interpolate between them, one per mechanism group:

- *Cohesion and bonding* (A, C, N, c^{α}): linear in the ice *volume* fraction ϕ_{ice} of Eq. (32) — ice cementation, the direct analogue of the `J2Erosion` rule $c = c^{\text{sed}} + S\,c^{\text{ice}}$, but measured per unit total volume rather than per unit pore volume.
- *Elasticity*: the (K, G) split described below, also in ϕ_{ice} — same mechanism, ice bonding the grain contacts.
- *Pore collapse* (κ_0, W, D_1): linear in the ice *saturation* S — ice filling the pores resists crushing, and how full the pores are is exactly S.
- *Friction and shape* (D, θ, L, ϕ, R, Q, ψ, D_2): properties of the sediment skeleton, taken from the thawed sublist and independent of both.

The ice volume fraction. What ice cementation delivers is set by how much ice there is per unit *total* volume, not per unit pore volume. Two integration points at $S = 1$ with porosities 0.2 and 0.6 hold three times different ice and are not equally bonded, yet S alone cannot tell them apart; with a depth-varying `ACE Porosity File` that difference is present in every column. The cohesion, bonding and elastic parameters therefore interpolate on

$$\phi_{\text{ice}} = \frac{S\,n}{n_{\text{ref}}}, \qquad n = n_0\left(1 - \frac{|\varepsilon_v^p|}{W}\right), \tag{32}$$

where n_0 is the local uncrushed porosity (the `ACE Porosity File` profile, or the constant `ACE Bulk Porosity`), $\varepsilon_v^p \le 0$ is the volumetric plastic strain on the crush curve of Eq. (19), so that n is the *current* porosity, and n_{ref} is the `Reference Porosity`, the porosity at which the two end-member sets were calibrated. ϕ_{ice} is clamped to $[0, 1]$, which binds only where the local porosity exceeds the calibration porosity.

Compaction enters as the *fraction* of the crushable volume consumed, $|\varepsilon_v^p|/W$, and not as the raw strain. The two readings coincide when $W = n_0$, which the ACE porosity override enforces by setting W to the profile value, but they are not the same in general: where $W > n_0$ the difference $n_0 + \varepsilon_v^p$ subtracts a volume larger than the pore space present, drives n negative, and reports no ice in material that is still frozen. Since $|\varepsilon_v^p|/W$ lies in $[0, 1]$ by the cap lock of Section 5, the form above keeps n in $[0, n_0]$ with no clamp of its own and with no tacit assumption that W and n_0 are the same quantity.

The normalization is what makes ϕ_{ice} an interpolation weight rather than a volume fraction in name only, and the choice of n_{ref} carries the content of the rule. Normalizing by the *local* porosity would collapse Eq. (32) to $\phi_{\text{ice}} = S\,(1 - |\varepsilon_v^p|/W)$ and cancel exactly the spatial porosity dependence the volume fraction was introduced to represent. Normalizing by a fixed calibration porosity keeps it: a point in a high-porosity layer at $S = 1$ reaches the frozen end member, one in a low-porosity layer at the same saturation falls short of it. There is no defensible default for n_{ref}, since it is a property of the calibration rather than of the material state, so the kernel requires it. When $n_0 = n_{\text{ref}}$ and nothing has crushed, $\phi_{\text{ice}} = S$ and the map reduces to the pre-2026-08 saturation map exactly.

Two choices keep Eq. (32) well posed. First, ε_v^p is read at the previous converged κ, so the parameters are *explicit* in the plastic state: they stay fixed across the integrator's substeps, which preserves the within-step ramping of Section 6 and leaves the drift correction and the $\Delta\kappa \le 0$ clamp on the ground they were verified on. Making the parameters implicit in κ would couple them to the return mapping itself. Second, the crush curve is evaluated on the pure-S map, which is the map its own parameters use. This breaks what would otherwise be a circular definition, since W is itself interpolated: a ϕ_{ice} built from $W(\phi_{\text{ice}})$ would be implicit, and self-reinforcing, because crushing lowers ϕ_{ice}, which lowers W, which raises $|\varepsilon_v^p|/W$, which lowers ϕ_{ice} again.

Pore collapse deliberately stays on S. The crush curve already carries the compaction history in κ; routing compaction back through the parameters that define that curve would double-count it, and would drive those parameters toward the thawed member, which is calibrated for loose saturated sediment, precisely when the material is densifying. That is the wrong direction, and it is the same separation of channels as the design constraint recorded at the end of this section.

Writing y_f and y_t for the frozen and thawed values of a parameter, and abbreviating $\phi \equiv \phi_\mathrm{ice}$, the rules are

$$\begin{aligned} y(\phi) &= (1-\phi)\, y_\mathrm{t} + \phi\, y_\mathrm{f} && \text{for } y \in \{A,\, C,\, N,\, c^\alpha\}, \\ K(\phi) &= K_\mathrm{t}^{1-\phi}\, K_\mathrm{f}^{\phi}, \qquad G(\phi) = G_\mathrm{t}^{1-\phi}\, G_\mathrm{f}^{\phi}, && \\ y(S) &= (1-S)\, y_\mathrm{t} + S\, y_\mathrm{f} && \text{for } y \in \{\kappa_0,\, W,\, D_1\}, \end{aligned} \tag{33}$$

while the friction and shape parameters $\{D,\, \theta,\, L,\, \phi,\, R,\, Q,\, \psi,\, D_2\}$ are held at their thawed (skeleton) values. Both moduli interpolate geometrically rather than linearly, for the reasons set out below. At the end members the map returns the frozen ($S = \phi_\mathrm{ice} = 1$) or thawed ($S = \phi_\mathrm{ice} = 0$) sublist exactly, so either limit is a standard `Cap` material. The saturation is clamped to $[0, 1]$ before use. Figure 1 shows the resulting surface at both end members.

The saturation itself comes from one of three sources: the `ACE_Ice_Saturation` field computed by the coupled thermal problem (`Have ACE Ice Saturation: true`), a time table (`Ice Saturation Time Values / Ice Saturation Values`), or the constant `Ice Saturation` (default 1.0, frozen). The saturation seen at the previous converged step is kept as the state `Ice_Saturation_State`, and the integrator ramps the parameters from the previous to the current saturation across its substeps — the within-step ramping of Section 6, without which thaw-under-load trajectories are hypersensitive to roundoff. Because the elastic strain, not the stress, is the state, the stored stress is re-expressed through the current stiffness when the moduli change ($\boldsymbol{\sigma}_n \leftarrow C^e(S) : C^e(S_\mathrm{old})^{-1} : \boldsymbol{\sigma}_n$; the finite-deformation path does this by construction). In the finite-deformation path the thermal stretch is removed from $\boldsymbol{F}$ before the mechanical response is evaluated.

Elasticity and ice saturation. `J2Erosion` scales the Young's modulus with ice saturation at fixed Poisson's ratio, which scales K and G together. The underlying mechanism is sound — permafrost stiffness is governed by what fills the pores and whether it bonds the grains, so the moduli are properly functions of S (temperature acts only through the freezing curve) — but the partition between the elastic constants must differ. The thaw transition replaces pore ice ($G \approx 3.5 \times 10^9$ Pa, $K \approx 9 \times 10^9$ Pa) with pore water ($G = 0$, $K \approx 2.2 \times 10^9$ Pa) inside a mineral skeleton ($K_s \approx 3.6 \times 10^{10}$ Pa for quartz), and the two moduli respond entirely differently:

- *Shear modulus:* the dominant, order-of-magnitude dependence. Frozen, ice welds the grain contacts and the composite G is of order 10^9 Pa; thawed, water carries no shear and the unbonded skeleton's contact stiffness is of order 10^7 Pa — a drop of roughly two orders of magnitude. This is why shear-wave velocity collapse is the standard geophysical signature of thaw.

- *Bulk modulus:* weak dependence, bounded below by the saturated mixture. While the pores remain water-filled, the undrained bulk stiffness follows the Reuss/Wood mixture of water and grains ($K \approx 5 \times 10^9$ Pa at porosity 0.4) — a factor-of-two drop from frozen, not orders of magnitude. This is the classical Gassmann fluid-substitution result, with the additional loss of G because ice, unlike water, had shear stiffness.

- *Poisson's ratio is then a prediction, not an input:* with $K \approx 5 \times 10^9$ Pa and G of order 10^7 Pa, $\nu \to 0.499$ — thawed saturated soil approaches the undrained incompressible limit, up from $\nu \approx 0.3$ frozen.

Scaling E at fixed ν instead predicts that the bulk stiffness of thawed saturated sediment collapses along with the shear stiffness, so thawed zones compact volumetrically under their own weight far more than the pore water permits — a plausible contributor to the large displacements that motivated `J2Erosion`'s displacement-norm failure criterion.

Accordingly, each end member carries its own (K, G) pair in place of modulus-multiplier fits. Both moduli interpolate geometrically (log-linearly) in ϕ_{ice}. For G the range spans two orders of magnitude, so linear interpolation overweights the frozen value; for K, review of the frozen-soil experimental literature (models, experiments and data examined July 2026) shows the bulk modulus following the same geometric trend with ice saturation, so the interpolation was changed from the linear form originally assumed to match the observed behavior. The Wood-mixture bound is calibration guidance for the thawed K, not a kernel computation: until pore pressure is resolved (the u–p extension below), the calibrated thawed member uses drained-skeleton moduli, since an undrained K would make thawed sediment bulk-stiffer than frozen soil. One numerical guard is built in: the thawed saturated limit $\nu \to 0.499$ volumetrically locks low-order hexahedra, so the effective Poisson ratio is capped (`Maximum Poissons Ratio`, default 0.45) by reducing K while preserving G — the trusted physics.

Design constraint: environmental softening must not flow through the hardening variable. When ice melts the physical cap moves toward the origin, which would violate the $\Delta\kappa \leq 0$ clamp if it were imposed on κ. As with temperature-dependent yield in metals, κ stores only the plastic compaction history while the surfaces are built from S-dependent parameters evaluated fresh each step. A corollary for calibration: because κ is plastic memory while $\kappa_0(S)$ is a parameter, a state integrated under frozen parameters and then read under thawed ones sits far inside the thawed crush curve — reinterpreted as heavily precompacted. End members should be calibrated with a common κ_0 unless that reinterpretation is intended.

Failure criteria. The cap model supplies constitutive failure indicators that replace the stress-based `J2Erosion` criteria with quantities that have model meaning, and ports the kinematic criteria unchanged. Each indicator is normalized so that the value 1 means exhaustion of its mechanism, and each criterion is enabled by a positive threshold (or limit) in the material input and disabled at the default 0:

- *Tension* (bit 0x01, `Tension Indicator Threshold`): the fraction of the zero-pressure shear capacity $F_f(0) - N$ lost to mean tension, $[\,(A - C) - F_f(I_1)\,]/(A - C - N)$, which reaches 1 where the shear surface pinches to the tensile apex $F_f(I_1) = N$ and the material has no remaining strength. This replaces the maximum-principal-stress check. The threshold *must* be set below 1: because the yield function squares $F_f - N$, it has a spurious second branch beyond the apex on which the apparent strength grows again with tension. The integrator never reaches that branch in practice precisely because the element fails first — the criterion is part of the model's domain of validity, not just an erosion trigger.

- *Backstress saturation* (bit 0x02, `Backstress Indicator Threshold`): $\sqrt{J_2^\alpha}/N$, equal to 1 when the kinematic hardening is exhausted ($G^\alpha = 0$: the yield surface has translated to the failure envelope and the shear response is perfectly plastic). This replaces the criterion that yield onset itself constitutes failure — the cap model can distinguish yielded-but-hardening material from exhausted material, which J_2 cannot.

- *Crush exhaustion* (bit 0x04, `Crush Indicator Threshold`): $|\varepsilon_v^p(\kappa)|/W$ on the crush curve, the quantity whose approach to 1 engages the cap lock (full pore collapse).

- *Plastic distortion* (bit 0x20, `Maximum Equivalent Plastic Strain`): $\varepsilon_{\mathrm{eqps}}$ against a user limit, the plastic complement of the total-distortion criterion below.

- *Total distortion* (bit 0x40, `Maximum Distortion`): `J2Erosion`'s strain limit ported unchanged — the norm of the isochoric right Cauchy–Green tensor, $\|J^{-2/3}F^TF\|/\sqrt{3}$, against a user limit. The measure is 1 at rest, so the limit must exceed 1 (the implementation rejects positive limits ≤ 1), and it includes the *elastic* stretch. That matters during thaw: as the ice bonding disappears the stiffness collapses by orders of magnitude and the material deforms enormously at nearly zero stress, so neither the stress-based indicators nor eqps grow — the total stretch is what detects the collapse, and removing such cells promptly is also what keeps the implicit solve well posed through the collapse transient. Peat raises the limit to at least $1 +$ peat and ocean exposure shrinks the excess over 1 by the eqps-limit weakening factor, both `J2Erosion` conventions. Requires finite deformation.

The backstress and crush indicators approach 1 only asymptotically, so their thresholds must also be below 1 (the implementation rejects thresholds ≥ 1). The two structural criteria — tilt angle (bit 0x08, `Critical Angle`) and displacement norm (bit 0x10, `Maximum Displacement`), which detect toppled or departed blocks — are not constitutive and carry over from `J2Erosion` unchanged, as does the element-death bookkeeping: tripped modes accumulate in the per-point `failure_modes` bitmask (an STK-backed element state), the per-cell `failure_state` diagnostic encodes per-mode trip counts in decimal digits, and a cell dies (`cell_death` = 1, with live propagation into the assembly's death-status vector so Newton stops assembling condemned cells mid-step) when a user-settable number of its integration points have failed (`Failed Integration Points For Death`, default: all). All criteria branch on Sacado values so the residual and Jacobian evaluations take identical paths, and the death states are consumed by the ACE solver and scatter machinery without modification.

The ACE environment. The kernel reads the same environmental data as `J2Erosion`. Depth profiles against z above mean sea level (`ACE Z Depth File` with `ACE Porosity File` and `ACE Peat File`, or the constant `ACE Bulk Porosity`) supply porosity and peat fraction at each integration point. The porosity does double duty: it overrides the crushable volume W — the pore space is the crushable volume, independent of what fills it — and it is the uncrushed porosity n_0 of the ice volume fraction, Eq. (32), which is where the depth variation of porosity reaches the cohesion, bonding and elastic parameters. Driving both roles from the same profile is also what makes $W = n_0$ at every point, the case in which the crushed fraction of Eq. (32) reduces to the plain difference $n_0 + \varepsilon_v^p$. With no porosity source the calibration porosity n_{ref} stands in for n_0, which makes $\phi_{\text{ice}} = S$ up to accumulated compaction. Peat raises the eqps limit by the factor $(1 + \text{peat})$, a provisional parity rule pending calibration. Sea level against time (`ACE Sea Level File` / `ACE Time File`) drives ocean-exposure weakening of submerged cells in the erodible sets: the cohesion by its factor (C rises toward A, shrinking the zero-pressure strength $A - C$ and the offset N while leaving the friction slope and asymptote intact), the moduli uniformly (so ν, and the stored-stress re-expression, are unaffected), and the eqps limit by its factor. Unlike the saturation, the weakening switches on instantaneously when a cell submerges (the `J2Erosion` convention); if waterline flips ever destabilize the drift correction the way unramped parameter jumps do, the submersion state should be ramped like the saturation.

Material input. In addition to the cap parameters of Section 10 (split per the map above between the `Frozen Parameters` and `Thawed Parameters` sublists, with the friction/shape parameters read from the thawed sublist), the `Permafrost` material block reads:

Input name	Default	Meaning
`K`, `G` (per sublist)	—	end-member bulk and shear m
`Maximum Poissons Ratio`	0.45	cap on the effective ν, preserv
`Have ACE Ice Saturation`	false	read S from the coupled ther
`Ice Saturation Time Values/Values`	—	time-table source for S
`Ice Saturation`	1.0	constant source for S
`Tension Indicator Threshold`	0 (off)	see failure criteria
`Backstress Indicator Threshold`	0 (off)	”
`Crush Indicator Threshold`	0 (off)	”
`Maximum Equivalent Plastic Strain`	0 (off)	”
`Critical Angle`	0 (off)	tilt, radians
`Maximum Displacement`	0 (off)	displacement norm
`Maximum Distortion`	0 (off)	total isochoric stretch, > 1
`Failed Integration Points For Death`	0 (= all)	per-cell death threshold
`Disable Erosion`	false	criteria never trip
`ACE Z Depth File`	—	z profile abscissae
`ACE Porosity File`, `ACE Peat File`	—	depth profiles
`ACE Bulk Porosity`	0 (off)	constant porosity (W overrid
`Reference Porosity`	*required*	n_{ref}, the porosity at which the bers were calibrated
`ACE Sea Level File`, `ACE Time File`	—	sea level against time
`ACE Cohesion Weakening Factor`	1	submerged-cell weakening
`ACE Stiffness Weakening Factor`	1	”
`ACE Eqps Limit Weakening Factor`	1	”
`Softening`, `Coherence Residual`, `Failure Strain`, `Failure Speed`	off	cohesion softening, applied on ice-fraction map; Section 5
`Finite Deformation`, `Substep Tolerance`, `Maximum Substeps`		as in Section 10

Per-criterion indicator fields, the failure bookkeeping states, the standard cap states, and the ice volume fraction (`Output Ice Volume Fraction`, the field `Ice_Volume_Fraction`) are available for Exodus output through the usual `Output ...` flags.

Calibration. The thawed end member comes from standard soil-mechanics cap parameters for silty/clay sediments plus the ACE porosity profiles (W); the frozen end member from refitting the experimental data behind the `J2Erosion` stiffness/strength fits to cap parameters, with friction (θ, D) taken from the thawed sediment so the frozen fit reduces to cohesion scaling — the original `J2Erosion` assumption, now in its natural place. The script `tests/LCM/ACE/MiniErosionPermafrost/calibrate_permafrost.py` performs this conversion: it evaluates the `J2Erosion` fit polynomials (which encode the experimental program) at the end members for a given porosity and deck scales and emits the `Frozen/Thawed`

`Parameters` sublists, including the mapping of the distortion strain limit to an eqps limit. The single-element verification decks of Section 9 double as calibration drivers.

Verification. The `Permafrost` battery extends the methodology of Section 9, with the reference implementation mirroring the saturation map, the parameter ramping, and the failure indicators. Trajectory tests: equivalence with `Cap` at $S = 1$, the full map at constant $S = 0.5$ (the ν cap active), and thaw under load ($S : 1 \to 0.3$ during confined compression) all match at machine precision. The porosity-profile test drives the two Gauss-point depth groups of a single element as independent material points against two reference runs. Five per-mode death tests verify each failure criterion: the indicator series match the reference at machine precision and the `failure_modes`/`failure_state`/`cell_death` series match the reference-predicted trip step exactly (the displacement criterion additionally checks the per-point indicator multiset against the analytic Gauss-point displacements). Finally, a MiniErosion-style coupled test runs `Permafrost` under the ACE thermo-mechanical solver: a deliberately tight displacement limit kills the settled upper cells early, thaw-driven failures follow as the bluff warms, and the run continues on the eroded mesh — exercising live death, the active-part update, and the workset rebuilds on the shared mesh.

Future extensions. In order of expected payoff: rate sensitivity by the Duvaut–Lions wrap of Regueiro and Foster [6], which regularizes any inviscid integrator and would represent storm-timescale ice creep; and pore-pressure/effective-stress coupling — the source paper Sun et al. [9] is itself a u–p poromechanics formulation, so the model is built for that extension if ACE ever resolves pore fluid. The u–p extension would also let the thawed bulk modulus take its undrained (Wood-bound) value instead of the drained-skeleton compromise above.

A Revision History of the Formulation

In 2026 the implementation was audited line-by-line against the sources listed in Section 1, treating the code as unreliable, and then verified against the independent reference of Section 9. Results produced with the pre-audit code differ from the current model in the following ways; all four defects are absent from every published validation case, which is why they survived.

1. *$\partial J_3/\partial \boldsymbol{\sigma}$ used the elementwise square $s_{ij}s_{ij}$* (no sum) instead of the matrix product $(\boldsymbol{s} \cdot \boldsymbol{s})_{ij}$ in Eq. (28). The two agree in principal axes and the term is multiplied by $\mathrm{d}\Gamma/\mathrm{d}J_3 \propto (1 - 1/\psi)$, so any run with $\psi = 1$ — including all of Sun et al. [9] — never exercised it. Runs with $\psi \neq 1$ received wrong plastic flow directions off the principal axes.

2. *X^g was built from F_f^g* (L, ϕ) instead of the yield F_f (D, θ) required by Regueiro and Foster [6, Eq. 14]; see Eq. (16). Invisible for associative parameters.

3. *The crush-curve derivative used the potential-surface cap position* (Q, L, ϕ) instead of the yield $X(\kappa)$ (R, D, θ); see Eq. (19). Invisible for associative parameters.

4. G^α *was unclamped*, allowing hardening reversal after overshoot of the back-stress limit surface; see Eq. (18). This is the likely root cause of the Newton divergence that led to the disabling of the original `CapModelPlasticity3D` regression tests years earlier.

In the same effort: the drift tolerance was made unit-system invariant ($10^{-12}E^2$); the $\Delta\kappa \leq 0$ clamp, present in the code but undocumented, was confirmed against the LAME reference and documented; the cap lock at full compaction was added from the reference; adaptive substepping was added; the implicit variant (`CapImplicit`) was removed; the model was renamed from `CapExplicit` to `CapModel` with material-database name `Cap`; and the earlier finite-deformation path (which skipped the predictor, disabled the normal-correction fallback, and updated $\boldsymbol{F}^p$ by an exponential map of the final flow direction) was replaced by the formulation of Section 7.

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